

SA-86:When Reasoning Collapses: A Depth-Aware Probe into LLM Reasoning(Student Abstract)

Azka Ikramullah¹, Abdul Majeed², Kyunghyun Lee², Seong Oun Hwang^{2*}

¹ Department of IT Convergence Engineering, Gachon University, South Korea

² Department of Computer Engineering, Gachon University, South Korea

*Corresponding Author

Research Problem	Research Questions	Methodology	Key Results
Problem: LLMs often perform well with shallow reasoning tasks, but performance collapses as reasoning depth increases. Why It Matters: Real-world tasks require multi-step reasoning. Chain-of-thought prompting provides an illusion of reasoning, but fails at deeper inference levels. Gap: Existing benchmarks don't evaluate depth, leading to missed insights about model brittleness.	RQ1: Do reasoning prompts hold up at deeper levels? RQ2: How does accuracy change across different reasoning depths? RQ3: Where does reasoning collapse occur? RQ4: How do open-source vs closed-source models behave?	Datasets: CLUTRR (Kinship Reasoning), ProofWriter (Logical Entailment) Models: GPT-5 (closed-source), LLaMA-3.1-8B, Qwen-2.5-7B, DeepSeek-32B, GLM-4.5-Air (open-source) Metrics: Accuracy, Collapse Depth (where performance drops below 50%) Method: Compare direct vs reasoning prompts across increasing depths.	GPT-5: Small reasoning gains at shallow depths, but collapse at depth 3. Open-Source Models: Inconsistent improvements at shallow depths, followed by rapid collapse. DeepSeek-32B and GLM-4.5-Air: No reasoning improvements at any depth.

Evaluation framework

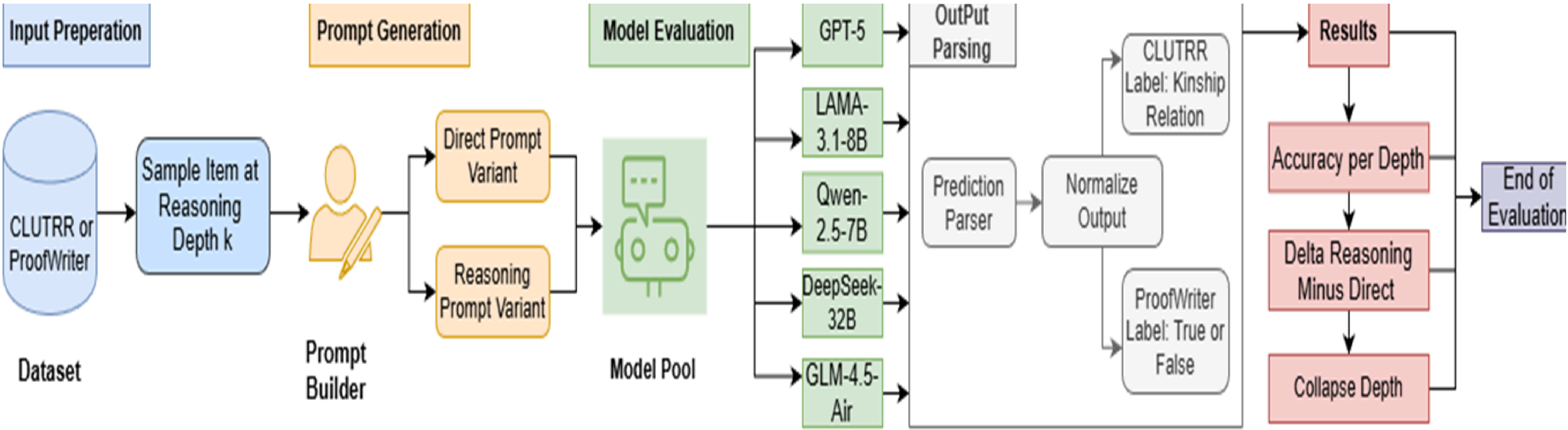
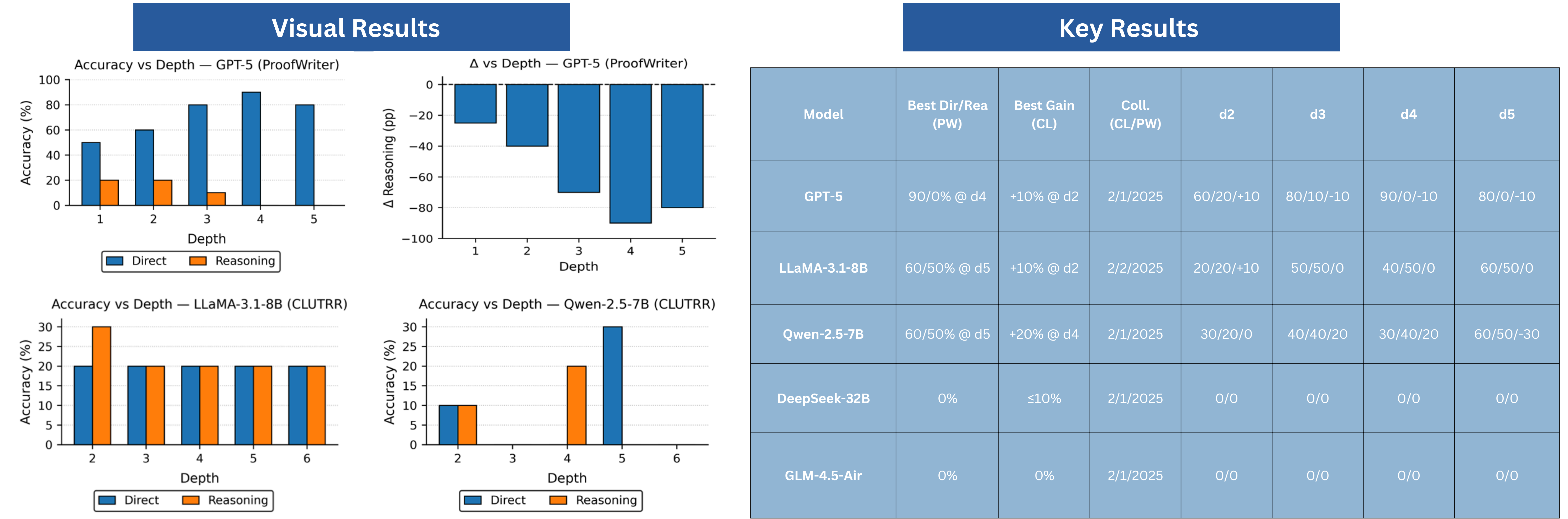


Figure 1: Evaluation framework that compares direct and reasoning prompts across controlled depths, tracking accuracy and introducing collapse depth as the robustness indicator.



Conclusion	Implications	Future Work
- Reasoning prompts offer limited benefits at shallow depths. - As the depth of reasoning increases, models' performance collapses, revealing depth-driven brittleness. - Collapse depth is a critical metric for assessing the robustness of reasoning in LLMs.	- Current benchmarks that do not measure reasoning beyond shallow depths tend to underestimate LLMs' limitations. - Depth-aware evaluations offer a better understanding of when reasoning breaks down and highlight model weaknesses that are not captured by conventional tasks.	- Extend depth-graded evaluations to broader reasoning benchmarks beyond CLUTRR and ProofWriter. - Investigate training strategies or architectural methods that could improve reasoning stability and mitigate collapse. - Assess depth-aware learning techniques to enhance performance on tasks requiring compositional and logical reasoning. - Develop frameworks that can handle complex multi-step reasoning tasks while maintaining consistent performance.